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Method for Predicting Dangerous Well Section of Gas-Induced Vibration Pipe String and Method for Preventing Failure

Abstract

The present disclosure provides a method for predicting a dangerous well section of a gas-induced vibration pipe string, including: respectively constructing mechanical models; coupling the mechanical models by adopting a finite element method based on field production parameters to obtain a full-well gas-induced vibration analysis model; selecting several nodes on the target gas extraction pipe string, and performing calculation according to the full-well gas-induced vibration analysis model to obtain dynamic characteristic parameters of the several nodes, the dynamic characteristic parameters including an average internal stress and an average external stress; performing calculation according to the average internal stress and the average external stress to obtain a safety factor n of each well section of the target gas extraction pipe string; and performing verification according to the safety factor n to obtain the dangerous well section of the target gas extraction pipe string.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to the field of oil and gas extraction technology, and in particular to a method for predicting a dangerous well section of a gas-induced vibration pipe string and a method for preventing a failure.

BACKGROUND

[0002] In the production process of “triple-high” gas wells (i.e., high-temperature, high-pressure and high-sulfur-content gas wells), the high-speed fluid in the target gas extraction pipe string may generate fluctuating pressure when passing through positions such as collars and nipples of the target gas extraction pipe string, causing the target gas extraction pipe string to expand and contract. At the same time, the deformation of the target gas extraction pipe string will react on the high-speed fluid, causing interaction and influence between the fluid and the target gas extraction pipe string. This fluid-solid coupling effect may cause vibration of the extraction pipe string in the gas well. If this vibration lasts for a long time, it will lead to problems such as natural gas leakage and shortened extraction pipe string life, causing serious economic losses to oilfield production. Therefore, in order to avoid the adverse situations, it is necessary to predict the dangerous well section and take some measures to prevent it from failing.

SUMMARY

[0003] In one aspect, the present disclosure provides a method for predicting a dangerous well section of a gas-induced vibration pipe string, including: respectively constructing a first mechanical model, a second mechanical model and a third mechanical model, the first mechanical model being a mechanical model of a target gas extraction pipe string, the second mechanical model being a mechanical model of a casing that sleeves the target gas extraction pipe string, the third mechanical model being a mechanical model of a fluid flowing in the target gas extraction pipe string; coupling the first mechanical model, the second mechanical model and the third mechanical model by adopting a finite element method based on field production parameters to obtain a full-well gas-induced vibration analysis model; selecting several nodes on the target gas extraction pipe string, and performing calculation according to the full-well gas-induced vibration analysis model to obtain dynamic characteristic parameters of the several nodes, the spacing between any two adjacent nodes of the several nodes being equal, the dynamic characteristic parameters including an average internal stress and an average external stress, the average internal stress being an average of internal stress values at the several nodes, the average external stress being an average of external stress values at the several nodes; performing calculation according to the average internal stress and the average external stress to obtain a safety factor n of each well section of the target gas extraction pipe string; and performing verification according to the safety factor n to obtain the dangerous well section of the target gas extraction pipe string.

[0004] In some embodiments, a method for establishing the full-well gas-induced vibration analysis model includes: **S201**: respectively performing discretization processing on the first mechanical model, the second mechanical model and the third mechanical model by using grid

units, and defining a coupling simulation area by using a node set; S202: setting an analysis parameter and selecting a gas flow model; S203: transmitting data through a fluid-solid coupling interface between each of the first mechanical model, the second mechanical model and the third mechanical model, and the gas flow model; S204: [0005] respectively solving a fluid control equation and a solid control equation of the gas flow model by using a partitioned method according to the analysis parameter in the coupling simulation area; and S205: repeating S203-S204 to obtain a correlation between fluid parameters and solid structure parameters, and constructing the full-well gas-induced vibration analysis model according to the correlation.

[0006] In some embodiments, a method for determining the grid units includes: respectively verifying grid independence and solution accuracy on at least one of the first mechanical model and the second mechanical model and on the third mechanical model to determine the optimal number of the grid units.

[0007] In some embodiments, the analysis parameter includes analysis time.

[0008] In some embodiments, the gas flow model includes a turbulence model.

[0009] In some embodiments, the performing calculation according to the average internal stress and the average external stress to obtain a safety factor n of each well section of the target gas extraction pipe string includes: performing calculation according to the average internal stress and the average external stress to obtain average stress $\sigma_{sub.s}$ of all well sections; determining maximum stress $\sigma_{sub.max}$ of each well section, the maximum stress $\sigma_{sub.max}$ being a maximum value of the internal stress values and the external stress values at all nodes of each well section; and calculating a direct ratio of the average stress $\sigma_{sub.s}$ to the maximum stress $\sigma_{sub.max}$ to obtain the safety factor n .

[0010] In some embodiments, the performing verification according to the safety factor n to obtain the dangerous well section of the target gas extraction pipe string includes: setting a standard safety coefficient $n_{sub.s}$ of the target gas extraction pipe string; and determining a well section with $n < n_{sub.s}$ as the dangerous well section.

[0011] In some embodiments, the standard safety coefficient $n_{sub.s}$ is 1.03-1.25.

[0012] In some embodiments, the standard safety coefficient $n_{sub.s}$ is 1.25.

[0013] In some embodiments, the dynamic characteristic parameters further include transverse and longitudinal deformation, transverse and longitudinal vibration speed, and three-dimensional bending moment.

[0014] In another aspect, the present disclosure provides a method for preventing a dangerous well section of a gas-included vibration pipe string from failing, including: determining the dangerous well section according to the method according to any one of the embodiments above; and mounting a plurality of stabilizers in the dangerous well section to reduce pipe string vibration.

[0015] In some embodiments, the number of all stabilizers of the plurality of stabilizers is 3-4.

[0016] In some embodiments, the spacing between any two adjacent stabilizers of the plurality of stabilizers is equal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1 illustrates a flowchart of a method for predicting a dangerous well section of a gas-induced vibration pipe string according to some embodiments.

[0018] FIG. 2 illustrates a flowchart of a method for predicting a dangerous well section of a gas-induced vibration pipe string according to some other embodiments.

[0019] FIG. 3 illustrates a flowchart of a method for predicting a dangerous well section of a gas-induced vibration pipe string according to some other embodiments.

[0020] FIG. 4 illustrates a flowchart of a method for predicting a dangerous well section of a gas-

induced vibration pipe string according to some other embodiments.

[0021] FIG. 5 illustrates a flowchart of a method for predicting a dangerous well section of a gas-induced vibration pipe string according to some other embodiments.

[0022] FIG. 6 illustrates a flowchart of a method for preventing a dangerous well section of a gas-included vibration pipe string from failing according to some embodiments.

[0023] FIG. 7A illustrates a schematic model of a casing, a target gas extraction pipe string and a fluid according to embodiment 1.

[0024] FIG. 7B illustrates a full well gas-induced vibration analysis model of a casing, target gas extraction pipe string and a fluid according to embodiment 1.

[0025] FIG. 8 illustrates a structural diagram of a target gas extraction pipe string according to embodiment 1.

[0026] FIG. 9 illustrates a diagram of positions of five selected equidistant nodes according to embodiment 1.

[0027] FIGS. 10A-10C illustrate charts of transverse and longitudinal deformation of a target gas extraction pipe string according to embodiment 1, where FIG. 10A is X-direction deformation, FIG. 10B is Z-direction deformation, and FIG. 10C is Y-direction deformation.

[0028] FIGS. 11A-11C illustrate charts of transverse and longitudinal bending moment time history of all well sections of a target gas extraction pipe string according to embodiment 1, where FIG. 11A is X-direction bending moment time history, FIG. 11B is Z-direction bending moment time history, and FIG. 11C is Y-direction bending moment time history.

[0029] FIGS. 12A-12E illustrate charts of transverse vibration speed of each equidistant node according to embodiment 1.

[0030] FIGS. 13A-13E illustrate charts of stress of each equidistant node according to embodiment 1.

[0031] FIG. 14 illustrates a chart of distribution of internal and average external stress of all well sections of a target gas extraction pipe string according to embodiment 1.

[0032] FIG. 15 illustrates a chart of internal and external safety coefficients of all well sections of a target gas extraction pipe string according to embodiment 1.

[0033] FIGS. 16A-16D illustrate charts of influences of the number of stabilizers on dynamic characteristic parameters of a target gas extraction pipe string according to embodiment 1, where FIG. 16A is influence on stress, FIG. 16B is influence on stress and fluctuation, FIG. 16C is influence on transverse deformation, and FIG. 16D is influence on vibration speed.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0034] In order to make the purposes, technical solutions and advantages of the present disclosure clearer, the present disclosure will be further described below with reference to the drawings. In the description of the present disclosure, it should be understood that the orientation or position relationships indicated by the terms “up”, “down”, “front”, “back”, “left”, “right”, “top”, “bottom”, “inside”, “outside” and the like are orientation or position relationships shown in the drawings, are only for the purposes of facilitating the description the present disclosure and simplifying the description instead of indicating or implying that the device or element referred to must have a specific orientation or be constructed and operated in a specific orientation, and thus cannot be understood as limitations on the present disclosure.

[0035] Gas-induced vibration refers to the phenomenon that the pipe string vibrates due to a significant fluid dynamic effect because high-speed gas flows through the pipe string and the gas flow rate changes greatly in the gas well extraction process. Such vibration may be amplified due to the interaction between the fluid and the pipe string (fluid-solid coupling effect), leading to structural stress concentration, fatigue damage, and even pipe string failure.

[0036] Pipe strings refer to long tubular structures used in oil and gas extraction, which are used to extract oil and gas from wells formed underground. In drilling and production processes, the pipe strings are important equipment responsible for transporting fluids, supporting well walls,

transmitting rotational or axial forces, and achieving various other functions. The types of pipe strings vary depending on their purpose and function, mainly including the following structures: casing, mounted in a drilled well to maintain well wall stability, prevent well wall collapse, and isolate different formations to prevent formation fluids from communicating with each other; oil pipe (gas pipe), located inside the casing to form a main passage for oil and gas to flow from the bottom of the well to the ground surface, with a diameter usually smaller than that of the casing.

[0037] The methods of research on gas-induced vibration in related technologies are mainly divided into three types. The first type is to establish an axial vibration, longitudinal vibration or fluid-sol coupling model according to the Hamiltonian principle, and solve it by using the Galerkin method, characteristic method, linear interpolation method, and transfer matrix method. The second type is to establish a finite element model of a single pipe string through commercial software, and analyze the influence of changes in factors such as internal pressure and axial force on the natural frequency. The third type is to establish an indoor experimental bench based on the principle of similarity, and study the transverse and longitudinal vibration of the pipe string based on the strain gauge testing technology.

[0038] Overall, the above research methods focus on the influence of different production and mechanical parameters on the natural frequency, and due to the random friction between the pipe string and the casing during vibration, conventional analytical methods are difficult to carry out calculation. It is difficult to simulate working conditions of high temperature and high pressure using indoor experiments, and only qualitative research can be conducted on the law of gas-induced vibration. The length of the pipe string that has been numerically simulated and analyzed still needs to be increased.

[0039] For the failure prevention of the gas-induced vibration pipe string, what are commonly used are vibration suppression and control techniques, which use auxiliary structures or devices to actively or passively intervene and adjust the pipe string vibration to reduce amplitude and stress and eliminate or weaken resonance or galloping phenomena. Stabilizers are the most important device. In common methods, arrangement of stabilizers does not consider the actual situation of the pipe string, but they are separately arranged at the wellhead and bottom of the well, or are arranged at equal distances throughout the entire section of the pipe string. Although such arrangement method has a certain vibration reduction effect, the amplitude reduction for the dangerous well section is limited.

[0040] On the other hand, there is relatively little research on the combination of an accurate method for predicting the dangerous well section of the gas-induced vibration pipe string and a method for preventing a failure in related technologies, and this technology is now a major gap.

[0041] In view of this, an embodiment of the present disclosure provides a method for predicting a dangerous well section of a gas-induced vibration pipe string. Referring to FIG. 1, the method includes S1-S5.

[0042] In S1, referring to FIG. 7B, a first mechanical model, a second mechanical model and a third mechanical model are respectively constructed.

[0043] It should be noted that the first mechanical model is a mechanical model of a target gas extraction pipe string, the second mechanical model is a mechanical model of a casing that sleeves the target gas extraction pipe string, and the third mechanical model is a mechanical model of a fluid flowing in the target gas extraction pipe string.

[0044] In S2, referring to FIG. 7B, the first mechanical model, the second mechanical model and the third mechanical model are coupled by adopting a finite element method based on field production parameters to obtain a full-well gas-induced vibration analysis model.

[0045] It should be noted that the field production parameters include the properties of the casing, the target gas extraction pipe string and the fluid, as well as the working environment parameters. Exemplarily, the field production parameters include the length of the target gas extraction pipe string, the inner diameter of the casing, the outer diameter of the target gas extraction pipe string,

the wall thickness of the casing and the target gas extraction pipe string, the shut-in pressure of the gas well, the pressure of the underground reservoir, the casing pressure of the casing, and the gas extraction amount of the gas well.

[0046] The coupling of mechanical models refers to the establishment of a unified multi-physical-field model that considers the interaction and influence between different mechanical models (such as fluid mechanical model and solid mechanical model), to solve complex mechanical problems. The commonly used methods for coupling mechanical models include two types, i.e., unidirectional coupling and bidirectional coupling. Unidirectional coupling refers to the influence of one physical field on another, while the other physical field has no effect on the one physical field. For example, in thermal stress calculation, only the influence of temperature on the structure is considered, while the influence of structural deformation on the temperature field is ignored. The calculation method of unidirectional coupling is to take the result of one physical field (such as pressure or temperature.) as the boundary condition or load of another physical field, and then solve the equations of the two physical fields separately. Its advantages are small calculation amount, simplicity, and ease of implementation, while the disadvantage is that it may ignore some important interaction effects. Bidirectional coupling refers to the interaction and influence between two physical fields. For example, in fluid-solid coupling calculation, the fluid generates pressure and viscous forces on the solid, while the solid generates deformation and speed on the fluid. There are two calculation methods for bidirectional coupling, including step-by-step calculation method and overall calculation method. The step-by-step calculation method is to solve two physical fields separately by their respective solvers, then exchange information through the interface, iterate repeatedly until convergence. The overall calculation method is to integrate the equations of two physical fields into a set of equations for overall coupling solution. The present disclosure preferably adopts the step-by-step calculation method of bidirectional coupling, which solves separately and exchanges data.

[0047] In S3, several nodes are selected on the target gas extraction pipe string, and calculation is performed according to the full-well gas-induced vibration analysis model to obtain dynamic characteristic parameters of the several nodes. The spacing between any two adjacent nodes of the several nodes are equal.

[0048] It should be noted that the dynamic characteristic parameters include an average internal stress and an average external stress. The average internal stress is an average of internal stress values at the several nodes. The average external stress is an average of external stress values at the several nodes.

[0049] In some examples, the dynamic characteristic parameters further include transverse and longitudinal deformation, transverse and longitudinal vibration speed, and three-dimensional bending moment.

[0050] The transverse and longitudinal deformation refers to the measurement of the deformation or displacement that occurs when a structure is subjected to external forces in the transverse direction (horizontal direction) and the longitudinal direction (vertical direction). In the dynamic analysis of the pipe string, the transverse and longitudinal deformation reflects the degree of elastic or plastic deformation of the pipe string in different directions. The transverse and longitudinal deformation can be verified through measurement and dynamic simulation. Dynamic simulation analysis can simulate the response of the pipe string under working load by applying Finite Element Analysis (FEA) to obtain transverse and longitudinal deformation data. The measurement of the actual deformation can be achieved by mounting strain gauges on the pipe string or using laser rangefinders.

[0051] The transverse and longitudinal vibration speed refers to the speed of a structure during transverse and longitudinal vibration, representing the variation of vibration displacement per unit time. This is an important parameter for analyzing the dynamic response characteristics of the pipe string under periodic forces. The transverse and longitudinal vibration speed can be verified

through vibration testing and dynamic simulation. Vibration testing typically uses accelerometers or velocimeters for measurement, and the data is analyzed for vibration characteristics through Fourier transform. Dynamic simulation analysis can predict the vibration speed of the pipe string under different conditions by simulating its vibration behavior through computer simulation.

[0052] The three-directional bending moment refers to the bending moment in three main directions (i.e., x-axis, y-axis and z-axis directions), which describes the rotational effect or torque experienced by the structure in these directions. It can also be verified through dynamic simulation and measurement. The measurement of the bending moment is achieved by attaching strain measuring devices (such as strain gauges) to the structure to measure the strain value at the designated position of the structure, and then converting the strain value into bending moment value for verification.

[0053] It should be understood that in the subsequent steps, the average internal stress and the average external stress can be used for calculating the safety factor of the target gas extraction pipe string, while other dynamic parameters (i.e., transverse and longitudinal deformation, transverse and longitudinal vibration speed, and three-directional bending moment) can be used for verifying the calculation process and results.

[0054] In S4, calculation is performed according to the average internal stress and the average external stress to obtain a safety factor n of each well section of the target gas extraction pipe string.

[0055] It should be understood that the average internal stress reflects the average stress state inside the target gas extraction pipe string after being subjected to force, while the average external stress shows the external stress situation of the target gas extraction pipe string. These two parameters work together to comprehensively evaluate the performance and stability of the structure in actual working environments.

[0056] In S5, verification is performed according to the safety factor n to obtain the dangerous well section of the target gas extraction pipe string.

[0057] The safety factor is an important indicator used for evaluating whether the target gas extraction pipe string can withstand the load during the production process without damage. The safety factor calculated according to the average internal stress and the average external stress can more accurately reflect the safety boundary of the target gas extraction pipe string. In the present disclosure, if the safety factor of a certain well section is lower than a predetermined safety factor, it means that the well section may not be able to reliably withstand the expected load under the current working conditions, and is therefore considered a dangerous well section.

[0058] To sum up, the method for predicting the dangerous well section of the gas-induced vibration pipe string provided in this embodiment of the present disclosure has established a full-well gas-induced vibration dynamic model for the first time, which can analyze the vibration response of the gas-induced pipe string under any production parameters and any wellbore structure. Based on the finite element analysis method and the fluid-solid coupling between the high-speed gas and the slender gas extraction pipe string, a full-well gas-induced pipe string vibration response model has been established. Compared with the related technologies, the present disclosure can quickly and comprehensively obtain the mechanical characteristics of the pipe string at any position, such as three-directional stress, transverse and longitudinal deformation of the pipe string, bending moment and vibration speed, can quickly predict the dangerous area of the pipe string, and has certain guiding significance for predicting the safety and fatigue life of the pipe string.

[0059] In some embodiments, referring to FIG. 2, a method for establishing the full-well gas-induced vibration analysis model includes S201-S205.

[0060] In S201, discretization processing is respectively performed on the first mechanical model, the second mechanical model and the third mechanical model by using grid units, and a coupling simulation area is defined by using a node set.

[0061] It should be understood that discretization refers to the practice of dividing a continuous physical domain (such as structural member and fluid domain) into smaller units or elements during finite element analysis. In this way, a continuous domain can be described with limited data points, allowing complex problems to be solved on a computer. In the present disclosure, the continuous structure of the pipe string is transformed into a finite number of grid units for numerical analysis of its dynamic behavior and structural response. This can accurately simulate and calculate the stress, deformation, and vibration characteristics of the pipe string under different working conditions.

[0062] The node set refers to a set of specific points defined in a numerical model, which are used for representing key positions in the model or to apply boundary conditions, loads, constraints, etc. In Finite Element Analysis (FEA) or other similar numerical simulation methods, the node set plays an important role as it helps to define the geometric shape, physical properties, and analysis area of the model. The organization of nodes in the set may influence the convenience, efficiency, and accuracy of analysis.

[0063] The coupling simulation area refers to an area where the fluid and the solid interact in the model. In this area, the mechanical behavior of the fluid (such as pressure and flow rate) directly influences the mechanical behavior of the solid structure (such as stress and deformation), and vice versa. Therefore, the coupling simulation area is a crucial part of the simulation, which ensures an accurate representation of the interaction between the fluid and the solid.

[0064] In **S202**, an analysis parameter is set and a gas flow model is selected.

[0065] It should be noted that the gas flow model refers to a theoretical model that uses mathematical equations and physical laws to describe and predict the movement state and behavior of gas under different conditions. According to the movement state of gas, the gas flow model can be divided into a laminar flow model and a turbulence model. The laminar flow model refers to the flow of gas particles moving along parallel streamlines without mixing or disturbance. The turbulence model refers to the flow of gas particles moving along irregular curves and experiencing strong mixing and disturbance.

[0066] In some examples, the gas flow model includes a turbulence model, since the turbulence model typically requires the introduction of turbulence parameters and turbulence equations to describe the average and fluctuating movement of gas. However, the research object in this embodiment of the present disclosure is high-speed gas in the production process of the “three-high” gas well, so choosing the turbulence model is more suitable.

[0067] In some examples, the analysis parameter includes analysis time. It should be understood that in the analysis of fluid-solid coupling, there may be significant differences in the time scale of the occurrence and evolution of different physical processes. Choosing the appropriate analysis time can ensure that the model correctly captures key physical phenomena, and within this time range, the simulation result can reflect actual physical behavior. Moreover, time also determines the selection of time step size and the calculation accuracy of dynamic response in the simulation process. Therefore, by setting the analysis time, the time accuracy and detail level of the simulation can be controlled, ensuring that the calculation can accurately capture the system's response at critical moments. This is crucial for understanding the behavior of the system under transient or dynamic conditions. Further, reasonable analysis time setting can help to optimize the use of calculation resources and avoid unnecessary calculation burden caused by excessive simulation time. In addition, it can also help to determine sufficient temporal resolution to accurately capture the physical phenomena of interest without losing important dynamic features.

[0068] In **S203**: data are transmitted through a fluid-solid coupling interface between each of the first mechanical model, the second mechanical model and the third mechanical model, and the gas flow model.

[0069] It should be noted that the data transmission at the fluid-solid coupling interface is bidirectional and refers to the transmission of CFD (Computational Fluid Dynamics) analysis and

calculation results (flow pressure, flow rate) to the solid structure for analysis, and the transmission of solid structure calculation results (displacement) to the fluid. The specific solution steps include defining the coupling analysis step size through CFD. The coupling analysis step size is the period between two consecutive joint simulation data exchanges. An Explicit analysis module imports the coupling step size through CFD. Analysis software will perform one increment (lock step) or multiple increments (sub-cycle) in each coupling step. For FSI analysis combining Explicit and CFD, Explicit typically uses sub-cycles, while CFD uses lock step behavior. Specifically, this step can be achieved by using software such as Phoenix, CFX, STAR-CD, and FLUENTQ.

[0070] In **S204**, a fluid control equation and a solid control equation of the gas flow model are respectively solved by using a partitioned method according to the analysis parameter in the coupling simulation area.

[0071] The partitioned method is a numerical method for solving fluid-solid coupling problems. It solves the control equations of fluid and solid separately by their respective solvers, and then exchanges information through interfaces. The fluid transfers pressure and viscous forces to the solid, while the solid feeds back deformation and speed to the fluid. The fluid control equation is a set of partial differential equations that describe the laws of fluid movement, usually including mass conservation equations, momentum conservation equations, energy conservation equations, etc. According to the characteristics such as fluid movement, phase change, compressibility and rotation, different physical models and numerical methods can be used to solve the fluid control equation. The solid control equation is a set of partial differential equations that describe the deformation laws of the solid, typically including equilibrium equations, geometric equations, constitutive equations, etc. Based on the material properties, stress state, strain state and other characteristics of the solid, different physical models and numerical methods can be used to solve the solid control equation.

[0072] In **S205**, **S203-S204** are repeated to obtain a correlation between fluid parameters and solid structure parameters, and the full-well gas-induced vibration analysis model is constructed according to the correlation. It should be noted that the interaction effect between the fluid and the structure cannot be calculated at once, but gradually approaches the true physical state by repeating the calculation steps of fluid-solid coupling. This method can help to accurately capture the interdependence between fluid dynamics and structural response, thus establishing an analysis model that can comprehensively reflect the phenomenon of gas-induced vibration in the entire well.

[0073] To sum up, the method for establishing the full-well gas-induced vibration analysis model provided in this embodiment of the present disclosure can perform discretization processing on the mechanical model by using grid units, which can transform a consecutive physical phenomenon into a numerical problem, thus enabling the complex fluid-solid coupling system to be simulated and analyzed on the computer. This process improves the flexibility and accuracy of model processing, allowing for detailed research on complex pipe string behavior. Choosing the appropriate analysis time and flow model helps to ensure the accuracy and practicality of simulation results. By transmitting data between the fluid mechanical model and the structural mechanical model, the models are allowed to share information at the coupling interface, enabling the models to accurately reflect the interaction between the fluid and the solid, thus improving the authenticity and reliability of the analysis. By iteratively optimizing the correlation between the fluid parameters and the solid structure parameters, it is helpful to construct a more accurate and comprehensive full-well gas-induced vibration analysis model, which can truly reflect the complex underground working environment and provide reliable scientific basis for predicting and preventing the dangerous well section of the gas-induced vibration pipe string.

[0074] It should be understood that the grid units divide the calculation domain into many small volumes or elements for numerical simulation and analysis of fluid dynamics problems. The selection of the number of grid units should be determined according to the characteristics of the flow problem and the target y^+ value. Generally, the more the grid units, the higher the calculation

accuracy, but it will also increase the demand for calculation time and storage space. Moreover, the number of grid units is not linearly related to the improvement of accuracy, but exponentially related to the demand for storage space. Therefore, reasonably selecting the number of grid units is an important step in model establishment. In the related technologies, grid sensitivity analysis is generally used to evaluate grid quality and convergence to determine.

[0075] Based on this, some embodiments of the present disclosure provide a method for determining grid units. Referring to FIG. 3, the method includes **S301**.

[0076] In **S301**, grid independence and solution accuracy are respectively verified on at least one of the first mechanical model and the second mechanical model and on the third mechanical model to determine the optimal number of the grid units. It should be understood that the first mechanical model and the second mechanical model are solid structure models, while the third mechanical model is a fluid model. In order to accurately grasp the characteristics of grid independence and solution accuracy in both solid structure model and fluid model, this step selects both the fluid model and at least one solid structure model.

[0077] It should be noted that verifying grid independence and solution accuracy refers to determining the convergence of calculation results under different numbers of grid units according to the calculation results of flow pressure under different numbers of grid units. By keeping the fluid and target gas extraction pipe string constraints unchanged, the grids can be gradually refined, and the pressure gradients under different numbers of grid units are calculated separately. By comparing with the average field pressure gradient, the optimal number of grid units can be selected. The casing plays a role in constraining the deformation boundary of the target gas extraction pipe string and the contact effect between the oil and the casing in this process. However, since the content of this part is not the focus of the present disclosure, it will not be repeated.

[0078] To sum up, the method for determining grid units provided in this embodiment of the present disclosure can determine an appropriate level of grid refinement by verifying the grid independence and solution accuracy of solid and fluid models, thus ensuring the accuracy of calculation results, helping to accurately simulate the behavior of the gas-induced vibration pipe string, and reducing errors caused by improper grid settings.

[0079] In some embodiments, referring to FIG. 4, the step of performing calculation according to the average internal stress and the average external stress to obtain a safety factor n of each well section of the target gas extraction pipe string includes **S401-S403**.

[0080] In **S401**, calculation is performed according to the average internal stress and the average external stress to obtain average stress $\sigma_{\text{sub.s}}$ of all well sections.

[0081] In **S402**, maximum stress $\sigma_{\text{sub.max}}$ of each well section is determined.

[0082] It should be noted that the maximum stress $\sigma_{\text{sub.max}}$ is a maximum value of the internal stress values and the external stress values at all nodes of each well section.

[0083] In **S403**, a direct ratio of the average stress $\sigma_{\text{sub.s}}$ to the maximum stress $\sigma_{\text{sub.max}}$ is calculated to obtain the safety factor n .

[0084] To sum up, in the method for determining the safety factor provided in this embodiment of the present disclosure, by comparing the average stress with the maximum stress, a smaller safety factor indicates that the maximum stress is closer to the average stress, which indicates the stress distribution is more uniform. On the contrary, a larger safety factor indicates that the maximum stress is much higher than the average stress, which indicates that the stress distribution is not uniform and there is a possible risk. This method for determining the safety factor helps to identify well sections with relatively uniform stress distribution, and also helps to identify dangerous well sections that may exceed the design standards.

[0085] In some embodiments, referring to FIG. 5, the step of performing verification according to the safety factor n to obtain the dangerous well section of the target gas extraction pipe string includes **S501-S502**.

[0086] In **S501**, a standard safety coefficient $n_{sub.s}$ of the target gas extraction pipe string is set.
[0087] It should be noted that according to the recommendations of the AQ2012-2007 Petroleum and Natural Gas Safety Regulations, the recommended design safety factor for the target gas extraction pipe string is 1.03-1.25. Therefore, in some examples, the standard safety factor $n_{sub.s}$ is 1.03-1.25.

[0088] On this basis, an upper limit is taken for the safety coefficient of a sulfur-containing natural gas well. Therefore, exemplarily, the standard safety factor $n_{sub.s}$ is 1.25.

[0089] In **S502**, a well section with $n < n_{sub.s}$ is determined as the dangerous well section.

[0090] To sum up, the method for predicting the dangerous well section of the gas-induced vibration pipe string provided in this embodiment of the present disclosure establishes a full-well gas-induced vibration dynamic model for the first time, which can analyze the vibration response of the gas-induced pipe string under any production parameters and wellbore structure. Based on the finite element analysis method and the fluid-solid coupling between high-speed gas and slender gas extraction pipe string, a full-well gas-induced pipe string vibration response model is established. Compared with the related technologies, the present disclosure can quickly and comprehensively obtain the mechanical characteristics of the pipe string at any position, such as the three-directional stress, transverse and longitudinal deformation, bending moment and vibration speed of the pipe string, can quickly predict the dangerous area of the pipe string, and has certain guiding significance for predicting the safety and fatigue life of the pipe string.

[0091] In another aspect, an embodiment of the present disclosure further provides a method for preventing a dangerous well section of a gas-included vibration pipe string from failing. Referring to FIG. 6, the method includes **S11-S12**.

[0092] In **S11**, the dangerous well section is determined according to the method predicting the dangerous well section of the gas-induced vibration pipe string according to any one of the embodiments above.

[0093] In **S12**, a plurality of stabilizers are mounted in the dangerous well section to reduce pipe string vibration.

[0094] In some examples, the number of all stabilizers of the plurality of stabilizers is 3-4.

[0095] In some examples, the spacing between any two adjacent stabilizers of the plurality of stabilizers is equal.

[0096] Stabilizer is a commonly used tool in oil and gas well drilling and completion operations. Its main function is to clamp the target pipeline (such as the target gas extraction pipe string in the present disclosure) to maintain its central position in the wellbore, so as to ensure the verticality and stability of equipment such as the pipe string in the well. It can to some extent reduce pipe string vibration caused by fluid dynamics.

[0097] To sum up, the method for preventing the dangerous well section of the gas-included vibration pipe string from failing provided in this embodiment of the present disclosure predicts the dangerous well section of the pipe string by analyzing the dynamic characteristics of the pipe string, and reduces the possibility of failure of the pipe string by mounting failure prevention tools in the dangerous well section. It should be understood that since the method for preventing the dangerous well section of the gas-included vibration pipe string from failing provided in this embodiment of the present disclosure determines the dangerous well section based on the method for predicting the dangerous well section of the gas-induced vibration pipe string provided in the embodiment of the present disclosure, the beneficial effects of the method for preventing the dangerous well section of the gas-included vibration pipe string from failing provided in this embodiment of the present disclosure include the beneficial effect of the method for predicting the dangerous well section of the gas-induced vibration pipe string provided in the embodiment of the present disclosure.

[0098] In order to objectively evaluate the technical effectiveness of the embodiments of the present disclosure, the methods provided in the present disclosure will be exemplarily described

below through specific embodiments.

Embodiment 1

[0099] This embodiment is carried out on the basis of embodiment 1 above. This embodiment provides a specific example of practical field application.

[0100] The size of the pipe string and the field production parameters in this embodiment are as shown in Table 1 below.

TABLE-US-00001 TABLE 1 Size of pipe string and field production parameters in this embodiment Length of pipe string: 5600 m Inner diameter of casing: 177.8 mm; outer diameter of casing: 88.9 mm; wall thickness: 6.5 mm Shut-in static pressure: 48.7 MPa Reservoir pressure: 73 MPa Casing pressure: 25 MPa Gas flow pressure gradient: 0.55 MPa/100 m Gas yield: 50.25*10⁴ m³/d Natural gas density: 220 kg/m³ Density of pipe string: 7850 kg/m³ Analysis time: 1 s

[0101] In S1, mechanical models of a casing, a target gas extraction pipe string and a fluid are respectively constructed aiming at the target gas extraction pipe string, as shown in FIG. 7B.

[0102] In S2, the mechanical models of the casing, the target gas extraction pipe string and the fluid are coupled based on a finite element analysis method and field production parameters to obtain a full-well gas-induced vibration analysis model. Before coupling, the mechanical models of the casing, the target gas extraction pipe string and the fluid are respectively discretized by using grid units. A method for determining the grid units includes verifying grid independence and solution accuracy from two aspects, i.e., the fluid and the target gas extraction pipe string, to determine the optimal number of the grid units.

[0103] Specifically, in this embodiment, the fluid and target gas extraction pipe string constraints are kept unchanged, the grids are gradually refined, the flow pressure calculation results under different grid numbers are compared, and the independence of results from the grids is determined, thus proving the convergence of numerical results. Taking the calculation of flow pressure results as an example, when the number of grids exceeds 168000, the numerical curves of the calculated pressure gradient tend to overlap. The calculated pressure gradient data are compared with the average field pressure gradient (0.55 MPa/100 m). The error rates are listed as shown in Table 2 below. When the number of grids exceeds 168000, the average error rate is only 4.25%, indicating that the verification of fluid grid independence is completed and the model calculation accuracy is high.

TABLE-US-00002 TABLE 2 Error rates of fluid pressure gradient

Number of grids	112020	168000	333600	373320	559980
Calculated pressure gradient	0.96	0.57	0.60	0.61	0.51 (MPa/100 m)
Error rate	41%	2%	5%	6%	4%

[0104] Referring to FIG. 8, five equidistant nodes on the target gas extraction pipe string are selected to analyze the average stress under five different grid numbers. It can be seen that when the number of grids exceeds 125440, the magnitude of node stress changes very little. The stress change rates of the equidistant nodes are as shown in Table 3 below. It can be seen that when the number of grids is between 87000 and 125440, the stress change rate of each node is the highest; when the number of grids is higher than 125440, the stress change rate of each node is very low, and the average change rate is only 7.04%, indicating that the numerical calculation converges and the error is small at this moment, so the verification of the grid independence of the target gas extraction pipe string is completed.

TABLE-US-00003 TABLE 3 Stress change rates of equidistant nodes

Analysis point	Point 1	Point 2	Point 3	Point 4	Point 5
87000-125440	11.22%	19.19%	49.19%	27.46%	33.17%
125440-209104	6.12%	10.96%	9.47%	2.12%	4.33%
209104-435000	9.47%	4.03%	3.55%	5.05%	6.37%
435000-627200	8.42%	16.78%	8.01%	1.12%	9.31%
Average	8.01%	10.59%	7.01%	2.75%	6.67%

[0105] From the above analysis, it can be seen that when the number of grids is more than 168000, the model calculation error rate and stress change rate are both low. Therefore, the number of grid units may be set to 168000.

[0106] In S3, referring to FIG. 9, several nodes on the target gas extraction pipe string are selected, and dynamic characteristic parameters of the several nodes are calculated through the full-well gas-induced vibration analysis model. The calculation results are as shown in FIGS. 10A to 14.

[0107] In S4, a safety factor of the target gas extraction pipe string is calculated according to the average internal stress and the average external stress at each node. The calculation results are as shown in FIG. 15.

[0108] In S5, the dangerous well section of the target gas extraction pipe string is verified according to the safety factor, and a well section with $n < 1.25$ is determined as the dangerous well section.

[0109] After determining the dangerous well section, the failure prevention of the dangerous well section may also be carried out according to the failure prevention method provided in the present disclosure. Specifically, 3-4 stabilizers are mounted at equal distances in the dangerous well section to reduce the pipe string vibration. In this experiment example, the influence of mounting 2-4 stabilizers at equal distances in the dangerous well section on stress, fluctuation, transverse deformation and transverse vibration speed are considered, and a comparison with the working condition without stabilizers is made. As the number of stabilizers increases, the stress value is decreased by 169.94%, 156.59%, and 225.43%, respectively, as shown in FIG. 16A. The stress fluctuation is decreased by 524.5%, 433.76%, and 1319.32%, respectively, as shown in FIG. 16B. After mounting 3-4 stabilizers at equal distances in the dangerous well section, the stress and its fluctuation, the transverse deformation and the transverse vibration speed of the target gas extraction pipe string are significantly reduced.

[0110] The basic principles, main features and advantages of the present disclosure are shown and described above. Those skilled in the art should understand that the present disclosure is not limited by the above embodiments. The above embodiments and the descriptions herein are only used for describing the principles of the present disclosure. Without departing from the spirit and scope of the present disclosure, various changes and improvements may be made to the present disclosure, all of which still fall within the scope of protection of the present disclosure. The scope of protection of the present disclosure shall be defined by the attached claims and their equivalents.

Claims

1. A method for predicting a dangerous well section of a gas-induced vibration pipe string, comprising: respectively constructing a first mechanical model, a second mechanical model and a third mechanical model, the first mechanical model being a mechanical model of a target gas extraction pipe string, the second mechanical model being a mechanical model of a casing that sleeves the target gas extraction pipe string, the third mechanical model being a mechanical model of a fluid flowing in the target gas extraction pipe string; coupling the first mechanical model, the second mechanical model and the third mechanical model by adopting a finite element method based on field production parameters to obtain a full-well gas-induced vibration analysis model; selecting several nodes on the target gas extraction pipe string, and performing calculation according to the full-well gas-induced vibration analysis model to obtain dynamic characteristic parameters of the several nodes, the spacing between any two adjacent nodes of the several nodes being equal, the dynamic characteristic parameters comprising an average internal stress and an average external stress, the average internal stress being an average of internal stress values at the several nodes, the average external stress being an average of external stress values at the several nodes; performing calculation according to the average internal stress and the average external stress to obtain a safety factor n of each well section of the target gas extraction pipe string; and performing verification according to the safety factor n to obtain the dangerous well section of the target gas extraction pipe string.

2. The method according to claim 1, wherein a method for establishing the full-well gas-induced

vibration analysis model comprises: **S201**: respectively performing discretization processing on the first mechanical model, the second mechanical model and the third mechanical model by using grid units, and defining a coupling simulation area by using a node set; **S202**: setting an analysis parameter and selecting a gas flow model; **S203**: transmitting data through a fluid-solid coupling interface between each of the first mechanical model, the second mechanical model and the third mechanical model, and the gas flow model; **S204**: respectively solving a fluid control equation and a solid control equation of the gas flow model by using a partitioned method according to the analysis parameter in the coupling simulation area; and **S205**: repeating **S203-S204** to obtain a correlation between fluid parameters and solid structure parameters, and constructing the full-well gas-induced vibration analysis model according to the correlation.

3. The method according to claim 2, wherein a method for determining the grid units comprises: respectively verifying grid independence and solution accuracy on at least one of the first mechanical model and the second mechanical model and on the third mechanical model to determine the optimal number of the grid units.

4. The method according to claim 3, wherein the analysis parameter comprises analysis time.

5. The method according to claim 3, wherein the gas flow model comprises a turbulence model.

6. The method according to claim 1, wherein the performing calculation according to the average internal stress and the average external stress to obtain a safety factor n of each well section of the target gas extraction pipe string comprises: performing calculation according to the average internal stress and the average external stress to obtain average stress $\sigma_{\text{sub.s}}$ of all well sections; determining maximum stress $\sigma_{\text{sub.max}}$ of each well section, the maximum stress $\sigma_{\text{sub.max}}$ being a maximum value of the internal stress values and the external stress values at all nodes of each well section; and calculating a direct ratio of the average stress $\sigma_{\text{sub.s}}$ to the maximum stress $\sigma_{\text{sub.max}}$ to obtain the safety factor n .

7. The method according to claim 6, wherein the performing verification according to the safety factor n to obtain the dangerous well section of the target gas extraction pipe string comprises: setting a standard safety coefficient $n_{\text{sub.s}}$ of the target gas extraction pipe string; and determining a well section with $n < n_{\text{sub.s}}$ as the dangerous well section.

8. The method according to claim 7, wherein the standard safety coefficient $n_{\text{sub.s}}$ is 1.03-1.25.

9. The method according to claim 8, wherein the standard safety coefficient $n_{\text{sub.s}}$ is 1.25.

10. The method according to claim 1, wherein the dynamic characteristic parameters further comprise transverse and longitudinal deformation, transverse and longitudinal vibration speed, and three-dimensional bending moment.

11. A method for preventing a dangerous well section of a gas-included vibration pipe string from failing, comprising: determining the dangerous well section according to the method according to claim 1; and mounting a plurality of stabilizers in the dangerous well section to reduce pipe string vibration.

12. The method according to claim 11, wherein the number of all stabilizers of the plurality of stabilizers is 3-4.
